

Stat 517
Homework # 5

1. Let $X_1, X_2, \dots, X_n$ be a random sample from a $\text{Gamma}(\alpha = 3, \beta = \theta)$ distribution, $0 < \theta < \infty$. Determine the MLE of θ .
2. Let $Y_1 < Y_2 < \dots < Y_n$ be the order statistics of a random sample from a distribution with pdf $f(x; \theta) = 1, \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}, -\infty < \theta < \infty$, zero elsewhere. This is a nonregular case. Show that every statistics $u(X_1, \dots, X_n)$ such that

$$Y_n - \frac{1}{2} \leq u(X_1, \dots, X_n) \leq Y_1 + \frac{1}{2}$$

is a mle of θ . In particular, $(4Y_1 + 2Y_n + 1)/6$, $(Y_1 + Y_n)/2$, and $(2Y_1 + 4Y_n - 1)/6$ are three such statistics. Thus, uniqueness is not, in general, a property of mles.

3. Suppose $X_1, \dots, X_n$ are iid with pdf $f(x; \theta) = 2x/\theta^2, 0 < x \leq \theta$, zero elsewhere. Note this is a nonregular case. Find:
 - (a) The mle of $\hat{\theta}$ for θ .
 - (b) The constant c so that $E(c\hat{\theta}) = \theta$.
 - (c) The mle for the median of the distribution. Show that it is a consistent estimator.
4. Suppose $X_1, X_2, \dots, X_n$ are iid with pdf $f(x; \theta) = \theta^{-1}e^{-x/\theta}, 0 < x < \infty$, zero elsewhere. Find the mle of $P(X \leq 2)$ and show that it is consistent.